Topic: Distinct real Eigenvalues

Question: Find the solution to the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{4t}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t}$$

C $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-4t}$$

D $$\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-4t}$$

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Solution: B

The coefficient matrix is

$$A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$$

and the determinant $|A - \lambda I|$ is

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 3 \\ 3 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (1 - \lambda)^2 - 9$$

$$|A - \lambda I| = \lambda^2 - 2\lambda - 8$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 - 2\lambda - 8 = 0$$

$$(\lambda + 2)(\lambda - 4) = 0$$

$$\lambda = -2, 4$$

Then for these Eigenvalues, $\lambda_1 = -2$ and $\lambda_2 = 4$, we find

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$$A-(-2)I=\begin{bmatrix}1-(-2) & 3\\ 3 & 1-(-2)\end{bmatrix}$$

$$A-(-2)I=\begin{bmatrix}3 & 3\\ 3 & 3\end{bmatrix}$$

and

$$A-4I=\begin{bmatrix}1-4 & 3\\ 3 & 1-4\end{bmatrix}$$

$$A-4I=\begin{bmatrix}-3 & 3\\ 3 & -3\end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix}3 & 3\\ 3 & 3\end{bmatrix} \qquad \begin{bmatrix}-3 & 3\\ 3 & -3\end{bmatrix}$$

$$\begin{bmatrix}3 & 3\\ 0 & 0\end{bmatrix} \qquad \begin{bmatrix}-3 & 3\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix} \qquad \begin{bmatrix}1 & -1\\ 0 & 0\end{bmatrix}$$

If we turn these back into systems of equations (in both cases we only need to consider the equation that we get from the first row of each matrix), we get

$$k_1+k_2=0 \qquad k_1-k_2=0$$

$$k_1=-k_2 \qquad k_1=k_2$$

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From the first system, we'll choose $k_2 = -1$, which results in $k_1 = 1$. And from the second system, we'll choose $k_2 = 1$, which results in $k_1 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1} e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t}$$

Therefore, the general solution to the homogeneous system will be

$$\overrightarrow{x} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{4t}$$

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Topic: Distinct real Eigenvalues

Question: Find the solution to the system of differential equations.

$$x_1' = 3x_1 + 2x_2$$

$$x_2' = 7x_1 - 2x_2$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{5t}$

B $\vec{x} = c_1 \begin{bmatrix} 2 \\ -7 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{-4t}$

C $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 2 \\ 7 \end{bmatrix} e^{-4t}$

D $\vec{x} = c_1 \begin{bmatrix} 2 \\ -7 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{5t}$

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Solution: D

The coefficient matrix is

$$A = \begin{bmatrix} 3 & 2 \\ 7 & -2 \end{bmatrix}$$

and the determinant $|A - \lambda I|$ is

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & 2 \\ 7 & -2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & 2 \\ 7 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & 2 \\ 7 & -2 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (3 - \lambda)(-2 - \lambda) - 7(2)$$

$$|A - \lambda I| = \lambda^2 - \lambda - 20$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 - \lambda - 20 = 0$$

$$(\lambda - 5)(\lambda + 4) = 0$$

$$\lambda = -4, 5$$

Then for these Eigenvalues, $\lambda_1 = -4$ and $\lambda_2 = 5$, we find

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$$A-(-4)I=\begin{bmatrix}3-(-4) & 2\\ 7 & -2-(-4)\end{bmatrix}$$

$$A-(-4)I=\begin{bmatrix}7 & 2\\ 7 & 2\end{bmatrix}$$

and

$$A-5I=\begin{bmatrix}3-5 & 2\\ 7 & -2-5\end{bmatrix}$$

$$A-5I=\begin{bmatrix}-2 & 2\\ 7 & -7\end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix}7 & 2\\ 7 & 2\end{bmatrix}$$

$$\begin{bmatrix}7 & 2\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & \frac{2}{7}\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}-2 & 2\\ 7 & -7\end{bmatrix}$$

$$\begin{bmatrix}1 & -1\\ 7 & -7\end{bmatrix}$$

$$\begin{bmatrix}1 & -1\\ 0 & 0\end{bmatrix}$$

If we turn these back into systems of equations (in both cases we only need to consider the equation that we get from the first row of each matrix), we get

$$k_1+\frac{2}{7}k_2=0$$

$$k_1-k_2=0$$

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$$k_1 = -\frac{2}{7}k_2 \qquad\qquad k_1 = k_2$$

From the first system, we'll choose $k_2 = -7$, which results in $k_1 = 2$. And from the second system, we'll choose $k_2 = 1$, which results in $k_1 = 1$.

$$\overrightarrow{k_1} = \begin{bmatrix} 2 \\ -7 \end{bmatrix} \qquad\qquad \overrightarrow{k_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} \qquad\qquad \overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 2 \\ -7 \end{bmatrix} e^{-4t} \qquad\qquad \overrightarrow{x_2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{5t}$$

Therefore, the general solution to the homogeneous system will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 2 \\ -7 \end{bmatrix} e^{-4t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{5t}$$

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Topic: Distinct real Eigenvalues

Question: Find the solution to the system of differential equations.

$$x_1' = x_1 + 6x_2$$

$$x_2' = 4x_1 - x_2$$

Answer choices:

A $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{5t}$

B $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{-5t}$

C $\vec{x} = c_1 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} te^{5t}$

D $\vec{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{5t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} te^{5t}$

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Solution: A

The coefficient matrix is

$$A = \begin{bmatrix} 1 & 6 \\ 4 & -1 \end{bmatrix}$$

and the determinant $|A - \lambda I|$ is

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 6 \\ 4 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 1 & 6 \\ 4 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 6 \\ 4 & -1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (1 - \lambda)(-1 - \lambda) - (6)(4)$$

$$|A - \lambda I| = \lambda^2 - 25$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 - 25 = 0$$

$$(\lambda - 5)(\lambda + 5) = 0$$

$$\lambda = -5, 5$$

Then for these Eigenvalues, $\lambda_1 = -5$ and $\lambda_2 = 5$, we find

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$$A-(-5)I=\begin{bmatrix}1-(-5) & 6\\ 4 & -1-(-5)\end{bmatrix}$$

$$A-(-5)I=\begin{bmatrix}6 & 6\\ 4 & 4\end{bmatrix}$$

and

$$A-5I=\begin{bmatrix}1-5 & 6\\ 4 & -1-5\end{bmatrix}$$

$$A-5I=\begin{bmatrix}-4 & 6\\ 4 & -6\end{bmatrix}$$

Put these two matrices into reduced row-echelon form.

$$\begin{bmatrix}6 & 6\\ 4 & 4\end{bmatrix} \qquad \begin{bmatrix}-4 & 6\\ 4 & -6\end{bmatrix}$$

$$\begin{bmatrix}1 & 1\\ 4 & 4\end{bmatrix} \qquad \begin{bmatrix}-4 & 6\\ 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix} \qquad \begin{bmatrix}1 & -\frac{3}{2}\\ 0 & 0\end{bmatrix}$$

If we turn these back into systems of equations (in both cases we only need to consider the equation that we get from the first row of each matrix), we get

$$k_1+k_2=0 \qquad k_1-\frac{3}{2}k_2=0$$

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$$k_1 = -k_2 \qquad k_1 = \frac{3}{2}k_2$$

From the first system, we'll choose $k_2 = -1$, which results in $k_1 = 1$. And from the second system, we'll choose $k_2 = 2$, which results in $k_1 = 3$.

$$\overrightarrow{k_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \overrightarrow{k_2} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Then the solutions to the system are

$$\overrightarrow{x_1} = \overrightarrow{k_1}e^{\lambda_1 t} \qquad \overrightarrow{x_2} = \overrightarrow{k_2}e^{\lambda_2 t}$$

$$\overrightarrow{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-5t} \qquad \overrightarrow{x_2} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{5t}$$

Therefore, the general solution to the homogeneous system will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-5t} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} e^{5t}$$

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